

HERMANN PAUWELLS RIVERA

Full-Stack Developer | Software Engineer

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PROFILE

Full-stack developer (TypeScript, React/Next.js, Node.js) in the final semester of a B.S. in Computer Science and Technology at Tecnológico de Monterrey; available full-time from January 2027. I build software people use: a live online store for a real business, real-time systems and APIs. Founder of three early-stage startups (AI agents orchestrated with TypeSafe's Jev, nutrition, AI video). I also work with C++/JUCE, DSP and machine learning.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Kotlin, C++, SQL, Swift, C#
Frontend & Apps: React, Next.js, Vite, Tailwind CSS, Jetpack Compose, Electron, Three.js
Backend & Data: Node.js, Express, FastAPI, Supabase, PostgreSQL, MySQL, SQLite
Tools: Git, GitHub Actions, TypeSafe's Jev, LLMs (OpenRouter), TensorFlow, FFmpeg, JUCE, Unity

EXPERIENCE

Full-Stack Developer (freelance), Juguetería El Arbolito Jun 2026 to present

- Built and shipped to production the online store and admin panel for a single-branch toy store with Next.js, TypeScript and Supabase (catalog, cart, checkout, backend).
- Integrated Mercado Pago payments and an LLM chatbot (OpenRouter); prepared the load of the 2,395-product catalog exported from the point of sale (Elevanta) and the online/in-store inventory sync (in progress).

STARTUPS (FOUNDER)

HIVEMIND Co-founder · team chat for people and AI agents · private pilot Sep 2026 to present

- Android (Kotlin/Jetpack Compose) and Windows (Electron) apps over a self-hosted ntfy server; HMAC-SHA256 signed messages, single-use invitations and 104 automated tests.
- Agent orchestration with TypeSafe's Jev (via OpenRouter): a Python dispatcher decides which AI agent takes each message or task (under US\$0.0001 per message, US\$0.50 daily cap), through our own gateway hosted on an Android phone.
- Jev-guided context compaction for long-running Claude Code bots (MIT fork of fast-jev-compaction with a token budget, decay and a lossless archive): ~57K tokens kept instead of ~229K, in 1.5 s.

Chécalo Founder · nutrition app for Mexico · in development Sep 2026 to present

- Scans barcodes (Open Food Facts) and computes Mexico's NOM-051 warning labels with deterministic rules checked against the official gazette (DOF); JavaScript PWA and Android app (Kotlin, CameraX, ML Kit).
- TypeSafe's Jev gives an explainable verdict and flags contradictions between ingredients and the nutrition table; every number comes from the rules. Per-product cached proxy: AI cost grows with the catalog, not with users.

recaps-studio Co-founder · AI narrated-video factory (novel → manhwa-style video) Sep 2026 to present

- Resumable Python pipeline with hash-based caching in SQLite: script, TTS with word-level timing (faster-whisper), shot planning, AI images with a vision judge, subtitles and FFmpeg/Remotion rendering.
- Human approval gates, a credit budget cap, automatic QA and pytest with mocked providers.

SELECTED PROJECTS

Lumina Reservations (WorkHub MTY) Academic team project · Node/Express, SSE, React 2026

- Implemented the reservation backend in Node/Express (services, repositories, controllers, migrations) and real-time occupancy updates over SSE.
- Built the floor map and the reservation flow in React/Vite (dashboard, filters, confirmation) with role-based access.

folk-park Personal project · C++20, JUCE 2026

- Wavetable VST3/Standalone synthesizer in C++20 and JUCE with deterministic MIDI generation.
- DSP architecture and automated tests; validated with pluginval strictness 5.

Other projects: HeatShield · CrediFiel Challenge (Datathon 2025) · Multi-agent warehouse simulation (RL) · MAWI · VisualControl (IoT) · Mildred Pierce · GateGenius (HackMTY 2025).

EDUCATION

B.S. in Computer Science and Technology (ITC), Tecnológico de Monterrey, Monterrey Campus 2021 to 2026
Expected graduation: December 2026.

AWARDS & ACTIVITIES

- Winner, Sustainability category, at Swift Challenge Fest 2025 (Tecnológico de Monterrey), with the HeatShield iOS app.
- SwiftTec Coding Community (2025): hackathons and innovation challenges.
- Music production and sound design (FL Studio).

Spanish (native) · English (C1)